30 · ThirtyMilestones MIS EXECUTIVE DASHBOARD

Technology & Scaling Plan

Prepared for Management · Decision Document · July 2026

> Our MIS Dashboard is live and running the entire company's operations from Google Sheets — Task Lists, Checklists and multiple offices, all tracked in one place. **Our data always stays inside Google Sheets** (safe and fully in our control). This document presents three options so that as the company and the number of sheets grow (from 3 today to 50+ tomorrow), the system continues to run with the same speed and reliability.

1. What is already delivered

- **Executive Dashboard** — KPIs, department performance and status breakdown; whole-company progress at a glance.
- **Task List & Checklist** — live from Google Sheets, showing each doer's work.
- **Done / Revise** — complete or reschedule a task from the dashboard; it writes straight back to Google Sheets.
- **Multi-Sheet support** — the Admin can connect new Google Sheets easily (a separate list for each office / work area).
- **Access control** — the Admin decides which sheet is visible to which doer.
- **Works across devices** — assign on one laptop and it appears identically on every device.
- **Clean view** — only pending work is shown; a task disappears the moment it is completed (the record stays in the sheet).
- **Speed** — server-side and browser caching make repeat loads fast.
- **Login & roles** — Admin / PC / Employee accounts.

2. Core principle — data stays in Google Sheets

The entire system is designed so that the **real data lives in Google Sheets** — because the data is safe there, it is backed up automatically, and the team can also open and edit it directly. Every option below respects this principle. The only difference between them is *how fast* and *at what scale* the data is served to the dashboard.

3. The scaling question

With 2–3 sheets today, everything runs well. When there are **50+ sheets** and many people using the system at once, the current "read the whole sheet every time" approach can become slow. There are three ways forward.

Option A — CURRENT

How: Google Sheets + Apps Script (with caching already added).

Speed: Acceptable (with cache), but slows under heavy load.

At 50 sheets: Apps Script limits (execution time, daily quotas) become a bottleneck.

Data: In Google Sheets. **Cost:** Free. **Effort:** Already built.

Verdict: Ideal for the current small scale. Not the long-term answer for a large company.

Option B — RECOMMENDED

How: Google Sheets (data unchanged) + the official **Google Sheets API** + Vercel serverless functions + a **cache layer**. Apps Script is retired.

Speed: Considerably faster — no cold starts, no 6-minute execution limit.

At 50 sheets: The cache layer absorbs both the load and the API rate limits.

Data: **Stays in Google Sheets** (100% the same). **Auth:** Proper login.

Cost: Free / low.

Effort: Medium (a one-time setup).

Verdict: The best balance while keeping data in Sheets — speed, scale and low cost. **The right choice for growth.**

Option C — ENTERPRISE

How: A true database (Supabase / Postgres) with Google Sheets kept as an **automatic backup/export**.

Speed: Instant, even across millions of rows. **At 50/500 sheets:** no difference.

Data: In the database (auto-copied to Sheets for backup). **Auth:** Best (row-level security).

Cost: Generous free tier, low thereafter. **Effort:** High (migration).

Verdict: For when the company becomes very large. Fastest and most scalable, but the data's primary home then becomes the database (with Sheets as backup).

4. Comparison at a glance

Criteria	A — Current (Apps Script)	B — Sheets API + Cache	C — Database (Supabase)
Speed	**Acceptable**	**Fast**	**Fastest**
50+ sheets	**Struggles**	**Comfortable**	**Comfortable**
Data in Google Sheets	**Yes**	**Yes**	**Backup / export**
Security / Login	**Basic**	**Proper**	**Best**
Cost	**Free**	**Free / Low**	**Free tier / Low**
Effort to build	**Done**	**Medium**	**High**
Fit for company scale	**No**	**Yes**	**Excellent**

5. Recommendation (phased plan)

Now (0–3 months): Option A — already built. Sufficient for the current scale, at zero cost.

Growth phase (as sheets/users increase): move to **Option B** — gaining speed and scale *while keeping the data in Google Sheets*. This is our **recommendation**.

Enterprise (very large company): Option C — a database with Google Sheets as automatic backup. Only when data and user volumes become very high.

In one line: the data always remains safe in Google Sheets. As we scale, Option B keeps the system fast and reliable even at 50+ sheets — at low cost.

6. Cost summary

All three options run at **no cost or very low cost**: Vercel — hosting (free/low) Google Sheets — free Sheets API — free Supabase — free tier

There is no large server or licence expense. A modest cost applies only as scale increases.

7. Decision required

- Shall we proceed with **Option B** (Sheets API + Cache) for growth? *(The data stays in Google Sheets.)*
- Or remain on **Option A** for now and review later?

Once approved, this can be implemented step by step without any interruption to daily use.